



BECS 11223
Fundamentals of
Programming

TELECOM BILLING SYSTEM



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BACKGROUND OF THE PROBLEM

Telecommunication companies provide various services such as prepaid and postpaid voice calls and mobile data packages. Managing customer billing, tracking usage, and handling recharges manually or with inefficient systems can lead to errors, delays, and poor customer experience.

Traditional billing processes may lack automation, making it difficult to maintain accurate records of user balances, usage history, and active packages. This creates inefficiencies in service delivery and customer management.

WHY THE PROBLEM IS IMPORTANT?

Accurate billing is critical in the telecom industry. Errors in billing can result in financial losses, customer dissatisfaction, and reduced trust in service providers.

With the increasing number of mobile users and data consumption, telecom systems must efficiently handle:

- Balance inquiry
- Package activation
- Usage tracking

A reliable billing system ensures transparency, efficiency, and improved customer satisfaction.

WHO IS AFFECTED BY IT?

- Telecom Companies: Face operational inefficiencies and revenue loss due to incorrect billing
- Customers: Experience incorrect charges, poor service, and inconvenience
- System Administrators: Struggle with managing large volumes of data manually

WHY THE SOFTWARE SOLUTION IS USEFUL?

A software-based telecom billing system automates core operations, reducing human error and improving efficiency. Benefits include:

- Fast and accurate billing calculations
- Easy balance management for prepaid users
- Tracking due dates of postpaid users
- Quick activation of voice and data packages
- Activate/Deactivate mobile numbers

This system will simulate real-world telecom operations in a simplified and efficient way.

PROPOSED SOLUTION

We propose to develop a Telecom Billing System in C that supports:

Key Features:

- Prepaid and postpaid account management
- Data package activation and tracking
- Balance inquiry and recharge functionality
- Activate/Deactivate connections

System Modules:

- User Account Management - Store user details (ID, name, type: prepaid/postpaid)
- Recharge System - Add credits to numbers
- Package Activation - Activate voice/data plans with validity
- Postpaid Billing – View the users which are due
- Activate/Deactivate connections

TECHNOLOGY AND TOOLS

Programming Language: **C**

IDE: **Code::Blocks**

Flowchart Design: **draw.io**

TASK ALLOCATION TO MEMBERS

Leader & Programmer	H.L.N.Chamindu
Backend logic & Flowchart	M.Haripirana
Documentation & Testing	V.Nithushi

PROJECT TIMELINE (4 WEEKS)

Week 1	Requirement analysis, system design, flowcharts
Week 2	Implement user management and basic structure
Week 3	Develop billing, recharge, and package modules
Week 4	Testing, debugging, final documentation

KEY MAIN FUNCTIONS

1. Balance Inquiry

- Display current balance (prepaid)
- Display due amount (postpaid)

2. Recharge

- Add credit to prepaid account
- Update due amount in postpaid account

3. Activate a Package

- Select voice/data package
- Deduct balance

4. Activate/Deactivate mobile numbers

- Perform system operations according to the activate state of the number

FLOWCHART

MOB_NUM	NIC	CUS_NAME	CON_TYPE
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↗ *USER_INFO_FILE*

MOB_NUM	AMOUNT	REM_DATA	REM_VOICE	BILL_DATE	STATE
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↗ *MOBILE_NUM_FILE*

INDEX	VALUE	DATA	VOICE
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↗ *PACKAGES_FILE*

[Click here](#) to see an enlarged version of the diagram

